THE GOOD-FACTORIZATION KERNEL CONDITION IS INDEPENDENT OF THE FACTORIZATION

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span; proof complete as written.

Allowed standard prerequisites (editorial).

- Domination of complete surjections. more-algebra-lemma-dominate-two-surjections, more-algebra-lemma-embed-map-Noetherian-complete-local-rings (cross-chapter, named). Produce the common dominating factorization S" -> S x_B S' used to reduce to comparable factorizations.
- Regular sequence lifting. algebra-lemma-grothendieck-regular-sequence, more-algebra-lemma-join-koszul-regular-sequences, more-algebra-lemma-noetherian-finite-all-equivalent (cross-chapter, named). Drive the comparable-factorization step: lifting regular sequences through flat maps and joining them.

Source reference. Original reference 00MB: focused-seeds/geometry/S9-dpa-ci-map-well-defined/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00MA: focused-seeds/geometry/S9-dpa-ci-map-well-defined/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00LM: focused-seeds/geometry/S9-dpa-ci-map-well-defined/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00MG: focused-seeds/geometry/S9-dpa-ci-map-well-defined/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00NR: focused-seeds/geometry/S9-dpa-ci-map-well-defined/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

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Source reference. Original reference 07NY: focused-seeds/geometry/S9-dpa-ci-map-well-defined/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 09Q8: focused-seeds/geometry/S9-dpa-ci-map-well-defined/ref-tags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07NN: focused-seeds/geometry/S9-dpa-ci-map-well-defined/ref-tags.txt, source lines 12–12. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0669: focused-seeds/geometry/S9-dpa-ci-map-well-defined/ref-tags.txt, source lines 13–13. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 09CC: focused-seeds/geometry/S9-dpa-ci-map-well-defined/ref-tags.txt, source lines 14–14. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $A \to B$ be a local homomorphism of Noetherian complete local rings. The following are equivalent*

1. *for some good factorization $A \to S \to B$ the kernel of $S \to B$ is generated by a regular sequence, and*
2. *for every good factorization $A \to S \to B$ the kernel of $S \to B$ is generated by a regular sequence.*

Proof. Let $A \to S \to B$ be a good factorization. As B is complete we obtain a factorization $A \to S^\wedge \to B$ where $S^\wedge$ is the completion of S. Note that this is also a good factorization: The ring map $S \to S^\wedge$ is flat (Algebra, Lemma 00MB), hence $A \to S^\wedge$ is flat. The ring $S^\wedge/\mathfrak{m}_A S^\wedge = (S/\mathfrak{m}_A S)^\wedge$ is regular since $S/\mathfrak{m}_A S$ is regular (More on Algebra, Lemma 07NY). Let $f_1, \ldots, f_r$ be a minimal sequence of generators of $\mathrm{Ker}(S \to B)$. We will use without further mention that an ideal in a Noetherian local ring is generated by a regular sequence if and only if any minimal set of generators is a regular sequence. Observe that $f_1, \ldots, f_r$ is a regular sequence in S if and only if $f_1, \ldots, f_r$ is a regular sequence in the completion $S^\wedge$ by Algebra, Lemma 00LM. Moreover, we have

$$S^\wedge/(f_1, \ldots, f_r)R^\wedge = (S/(f_1, \ldots, f_n))^\wedge = B^\wedge = B$$

because B is $\mathfrak{m}_B$-adically complete (first equality by Algebra, Lemma 00MA). Thus the kernel of $S \to B$ is generated by a regular sequence if and only if the kernel

of $S^\wedge \to B$ is generated by a regular sequence. Hence it suffices to consider good factorizations where S is complete.

Assume we have two factorizations $A \to S \to B$ and $A \to S' \to B$ with S and S' complete. By More on Algebra, Lemma 09Q8 the ring $S \times_B S'$ is a Noetherian complete local ring. Hence, using More on Algebra, Lemma 07NN we can choose a good factorization $A \to S'' \to S \times_B S'$ with S'' complete. Thus it suffices to show: If $A \to S' \to S \to B$ are comparable good factorizations, then $\mathrm{Ker}(S \to B)$ is generated by a regular sequence if and only if $\mathrm{Ker}(S' \to B)$ is generated by a regular sequence.

Let $A \to S' \to S \to B$ be comparable good factorizations. First, since $S'/\mathfrak{m}_R S' \to S/\mathfrak{m}_R S$ is a surjection of regular local rings, the kernel is generated by a regular sequence $\overline{x}_1, \ldots, \overline{x}_c \in \mathfrak{m}_{S'}/\mathfrak{m}_R S'$ which can be extended to a regular system of parameters for the regular local ring $S'/\mathfrak{m}_R S'$, see (Algebra, Lemma 00NR). Set $I = \mathrm{Ker}(S' \to S)$. By flatness of S over R we have

$$I/\mathfrak{m}_R I = \mathrm{Ker}(S'/\mathfrak{m}_R S' \to S/\mathfrak{m}_R S) = (\overline{x}_1, \ldots, \overline{x}_c).$$

Choose lifts $x_1, \ldots, x_c \in I$. These lifts form a regular sequence generating I as S' is flat over R, see Algebra, Lemma 00MG.

We conclude that if also $\mathrm{Ker}(S \to B)$ is generated by a regular sequence, then so is $\mathrm{Ker}(S' \to B)$, see More on Algebra, Lemmas 0669 and 09CC.

Conversely, assume that $J = \mathrm{Ker}(S' \to B)$ is generated by a regular sequence. Because the generators $x_1, \ldots, x_c$ of I map to linearly independent elements of $\mathfrak{m}_{S'}/\mathfrak{m}_{S'}^2$ we see that $I/\mathfrak{m}_{S'} I \to J/\mathfrak{m}_{S'} J$ is injective. Hence there exists a minimal system of generators $x_1, \ldots, x_c, y_1, \ldots, y_d$ for J. Then $x_1, \ldots, x_c, y_1, \ldots, y_d$ is a regular sequence and it follows that the images of $y_1, \ldots, y_d$ in S form a regular sequence generating $\mathrm{Ker}(S \to B)$. This finishes the proof of the lemma. $\square$